JianghuSect 2 (JS2-v2.6.0) — Deep-Dive Feature & Logic Audit

System #02 (Part 3 of 3 — Exhaustive Second-Pass Sweep): Executive & Frontier Autonomy, Tactical Combat Brain, Lineage/Mentorship Lifecycle & Hidden Skill-Key Typos

- **Target Codebase**: Js2 new (SIM.PATCH_VERSION = '2.6.0', SIM.SAVE_SCHEMA_VERSION = '2.6.0-living-chronicle-and-reporting')
- **Target Readers**: Autonomous Coding Models (**GPT** & **Gemini** Systems Architects / AI Engineers)
- **Prerequisite Companion Specs**:
  - 02A_JS2_Deep_Audit_NPC_Brain_Part1_Epistemic_Knowledge_and_Information_Asymmetry
  - 02B_JS2_Deep_Audit_NPC_Brain_Part2_Orphaned_Systems_Utility_Math_and_Goal_Loops
- **Modules Audited in This Second-Pass Sweep**: js/sect.js, js/combat.js, js/expedition.js, js/bottleneck.js, js/external_market.js, js/visitors_clues.js, js/ai.js
- **Why Part 3 Exists**: Parts 02A and 02B audited individual disciple perception (COG.perceive), daily AI.ACTS, and cognition.js goals. This exhaustive second pass audits **every remaining decision surface where NPCs act (or fail to act)** across **Executive/Frontier Politics**, **Real-Time Combat AI**, **Marriage/Lineage/Mentorship/Defection**, and **Recruitment/Appointment Evaluators**.

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9. [BRAIN-CMD-08] Player-Command UI Creep Violating Pure Autonomous Simulation (v3.3–v4.1 Systems Missing NPC Executive Brains)

9.1 Core Simulation Rule Violation

JianghuSect 2's foundational rule states: **Everything in the world happens naturally through NPC decisions; the player cannot command NPCs or steer sect operations directly (only edit stats or spawn items).** However, multiple subsystems added in v2.0.18–v4.1.0 were wired to **data-do="..." UI command buttons** while **never being added to SECT.masterDecision, SECT.hallHeads, or SECT.runOfficerJurisdictionSingle**!

9.2 Complete Matrix of Executive & Frontier Decisions Missing Autonomous NPC Drivers

Subsystem / Decision	Current Code Anchor	Why It Never Happens (or Stagnates) Without Player Clicking UI	Required Autonomous NPC Decision Owner
1. Sect Charter & Doctrine (`enactSectCharter`)	`js/sect.js` (`SECT.CHARTER_DOCTRINE`)	Initialized permanently to `'CHARTER_MERITOCRACY'`. Even if a greedy merchant, hermit scholar, or ruthless warlord becomes Sect Master via `SECT.succession` / `leadership` duel, **no NPC ever calls `enactSectCharter`!**	**Sect Master** (`SECT.masterDecision`, cooldown 180d, when Master's `styleOf(m)` diverges strongly from active Charter & `stones >= reserve + 250`).
2. Founding the 8 Sacred Peaks (`SECT.PEAKS.foundPeak`)	`js/sect.js` (`SECT.PEAKS`)	All 8 Peak Sanctuaries (`144` stations on map!) start `founded = false` in mist. Check that `SECT.masterDecision` / `ELDER_STATUS` autonomously founds matching Peaks when population/elders/stones meet criteria without player clicking UI!	**Sect Master + Qualified Stage-5+ Elder** matching the Peak's Dao affinity when `pop >= 22` and `stones >= reserve + peakCost`.

Subsystem / Decision	Current Code Anchor	Why It Never Happens (or Stagnates) Without Player Clicking UI	Required Autonomous NPC Decision Owner
3. 6-Officer Strategic Council (`conveneCouncil` & `resolveCouncilAgenda`)	`js/sect.js` (`SECT.STRATEGIC_COUNCIL`)	Exposed via `data-do="convene-strategic-council"`. Never convened or voted on autonomously by the 6 Grand Officers + Sect Master!	**Seasonal Council (`W.day % 90 === 15`)**: 6 Officers + Master score `'FORTIFY_LOGISTICS'`, `'DIPLOMATIC_OFFENSIVE'`, and `'GRAND_WAR_DRILL'` by their own `HC` traits & world threat!
4. Frontier Outposts & Supply Convoys (`establishOutpost`, `dispatchSupplyConvoy`)	`js/sect.js` (`SECT.OUTPOSTS_LOGISTICS`)	Exposed via `data-do="establish-outpost"`. `tickOutpostLogistics` drains supplies daily, **but no officer autonomously establishes camps on owned resource nodes or dispatches resupply convoys before morale collapses!**	**`quartermaster` / `mission` / `steward`** during `COG.performDuty` when a player-sect `resourceNode` lacks an outpost or `grainStock < 8`.
5. World Anomalies, Resource Node Claims, Garrisons & Vassal Distress (`expedition.js`)	`departWorldAnomaly`, `departResourceClaim`, `assignNodeGarrison`, `respondVassalDistress`, `marshalStrikeForce`	`X.daily` spawns `WORLD_ANOMALY` (`line 672`) and `SECRET_REALM` (`line 664`), and rivals auto-claim them on expiry (`line 669`)—verify and ensure **`mission` / `commander` / `envoy` / `X.volunteer` autonomously**	**`commander` + `mission` + `envoy` + `Sect Master`** weighted by `policy.expansion`, `policy.risk`, `courage`, `empathy`, and `partyPower vs rivalPower`.

Subsystem / Decision	Current Code Anchor	Why It Never Happens (or Stagnates) Without Player Clicking UI	Required Autonomous NPC Decision Owner
		dispatch parties to contest open anomalies, claim nodes, garrison owned nodes, answer `vassalDistressCalls` (based on Master's `loyalty`/`empathy` vs `caution`), and launch `sect_siege` only under extreme blood-vendetta + hawkish Master!	
6. Defector Pursuit Ultimatums (`resolveDefectorUltimatum`)	`js/sect.js` (`tickDefectorPursuit`)	Issues `W.sect.activePursuitUltimatum` (`status: 'PENDING'`), waiting for UI button `data-do="resolve-defector-ultimatum"`!	**Sect Master + `warden` / `envoy`** within 2 days: righteous/loyal/protective Master chooses `'PROTECT_AND_FIGHT'`; cautious/wealthy Master chooses `'COMPENSATE_STONES'`.
7. Specialist Facilities, Barrier Element, Gate Beasts & Closed-Door Toggle	`upgradeSpecialistFacility`, `setBarrierElement`, `assignGateBeast`, `autoTriggerClosedDoorOnBottleneck`	`SECT.upgradeBuilding` (`sect.js:619`) only considers 16 standard buildings—it **never upgrades `mech_workshop`, `beast_menagerie`, or `harmonic_pavilio`**	**`steward`/`master`** upgrades specialist facilities when path disciples >= 3; **`arraymaster`** tunes `BarrierElement`; **`beastmaster`** assigns `gateBeasts`; NPCs choose

Subsystem / Decision	Current Code Anchor	Why It Never Happens (or Stagnates) Without Player Clicking UI	Required Autonomous NPC Decision Owner
		n! `autoTriggerClosedDoorOnBottleneck` defaults to `false` waiting for UI toggle!	`enterClosedDoorSeclusion` themselves!

> **Rule for UI Cleanup / Preservation**: Keep diagnostic readouts in `UI`, and if sandbox/god-mode triggers remain, clearly label them as `[อำนาจฟ้าดิน/ผู้สังเกตการณ์]` while ensuring **100% of these systems run autonomously via NPC personalities when the player never touches a single governance button!**

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10. `[BRAIN-SOC-09]` Lifecycle, Marriage/Lineage, Mentorship Split & Defection Blind Spots

10.1 Exact Code Anchors

- `js/ai.js`: `AI.updateBond` (lines 744–762), `AI.kill` dying-wish Qi transmission (lines 887–890)
- `js/sect.js`: `SECT.takePersonalDisciple` (lines 688–693), `SECT.checkChallenge` (lines 902–911), `SECT.checkDesertion` (lines 912–926)
- `js/expedition.js`: `X.recruitResult` (line 244)
- `js/combat.js`: `C.tournamentEnd` (line 901)

10.2 Root-Cause Logic & Continuity Bugs

Defect 9A — Married Couples (`spouse`) Never Conceive Children & Are Disconnected From `companionId` (`TAOIST_COMPANION`)

1. **Split Companion State (`ai.js:750–760` vs `ai.js:823` & `combat.js:527`)**:
   - When two disciples court organically via `A.pair` (`date`) until `r.l > 85 && r.a > 70`, `AI.updateBond` sets `r.bond = 'คู่ครอง'` and `x.h.life.family.spouse = other.id`.
   - **It NEVER sets `x.companionId = other.id` or `RED_THREAD`!**
   - Consequently, organically married spouses **do NOT receive** dual-cultivation `+1.2` Qi (`ai.js:823` checks `n.companionId === o.id`), **do NOT receive**

`+2.5` bodyguard intercept priority (`combat.js:414` checks `al.companionId === d.id`), and **do NOT receive** `+15%` Red-Thread combat synergy!

2. **Zero Second-Generation Childbirth in Multi-Century Sects**:
   - `ai.js:625` checks `n.h.life.family.children`, and `n.parentIds` is referenced in kin checks, **yet zero code in the simulation ever allows married opposite-sex (or spirit-blessed) couples in the sect to conceive and raise children**!
   - Across a 300-year simulation, **zero sect-born second-generation disciples ever appear**: hereditary bloodlines (`TR_BL_*`), parent-child mentorship, family clans inside peaks, and `ANCESTRAL_RITUAL` three-generation lineage bonuses never emerge naturally from NPC romance!

Defect 9B — Mentorship Field Split (`mentor` / `family.disciples` vs `mentorId` / `apprenticeIds`) Breaks Dying-Wish Qi Transmission!

Compare how personal disciples are recorded across files:

- `sect.js:690` (`SECT.takePersonalDisciple`), `expedition.js:244` (`X.recruitResult`), and `combat.js:901` (`C.tournamentEnd`) write:

```JavaScript
d.mentor = m.id;
m.h.life.family.disciples.push(d.id);
d.h.life.family.master = m.id;
```

  They **DO NOT write `d.mentorId = m.id` or `m.apprenticeIds.push(d.id)`**!

- Now inspect `AI.kill` (`ai.js:887`):

```JavaScript
if (n.cult && !n.qiTransmitted && Array.isArray(n.apprenticeIds) &&
n.apprenticeIds.length > 0 && g.SECT && g.SECT.MENTORSHIP &&
g.SECT.MENTORSHIP.transmitQi)
```

  And `combat.js:814` (`BLACK_THREAD` vengeance when a master or disciple is slain!):

```JavaScript
const isClose = n.companionId === victim.id || n.mentorId === victim.id ||
victim.mentorId === n.id || ...
```

- **Broken Chain Reaction**: Disciples recruited by the Sect Master (`takePersonalDisciple`), Tournament Champions (`tournamentEnd`), and Expedition Recruits (`recruitResult`) have `apprenticeIds === undefined` and `mentorId === undefined`—so **dying Masters fail to transmit their lifelong Qi to them**, and **`BLACK_THREAD` blood-vengeance fails to trigger if their Master is murdered**!

Defect 9C — High-Stage Defectors & `TR_IN_FREE_SPIRIT` Evaporate Instead of Joining Rival Sects or Founding Splinter Groups

- `data_core.js` defines `TR_IN_FREE_SPIRIT` with `behaviorMods.decisionBias: { foundOwnSect: 30 }`, and `sect.js:912` (`checkDesertion`) marks dissatisfied disciples `n.deserting = true; n.exiled = true`.
- Instead of deserting elders/prodigies **joining an ideologically compatible Jianghu sect (`W.world.allSects figures`)**—where they could later return as a **rival envoy or raid commander** with a grudge!—`SIM.retire` deletes them from existence (and, as found in `CTX-03`, even donates their private manuals to our sect library!).

10.3 Remediation Blueprint

1. **Unify `spouse` <-> `companionId` & Add Bounded Generational Childbirth**:
   - When `AI.updateBond` marries `x` and `b`, set `x.companionId = b.id; b.companionId = x.id;` and bind `RED_THREAD` via `SECT.DESTINY_KARMA.bindThread`.
   - In `SECT.monthly`, married living couples (`age 18..48` or scaled by `lifespanMax`, `!away`, `freeHousingBeds(W) >= 1`, `children.length < 2`) have a bounded monthly chance to welcome a sect-born youth (`parentIds: [a.id, b.id]`), inheriting blended `HC` traits / `root` elements / rare `TR_BL_*` bloodlines!
2. **Canonical Helper `SECT.MENTORSHIP.linkMasterDisciple(W, master, disciple)`**:
   - Synchronize `disciple.mentor = disciple.mentorId = master.id`, `disciple.h.life.family.master = master.id`, `master.apprenticeIds`, and `master.h.life.family.disciples` in **one** place, and allow lonely Stage-5+ Elders (`teaching > 52`, `apprenticeIds.length < 2`) to autonomously take promising inner/core disciples!
3. **Persist Deserted / Exiled Notable Cultivators into `W.world`**:
   - When a Stage-3+ cultivator deserts or is exiled, transfer their private gear/manuals *with* them into a rival `W.world.allSects` roster (`figures`) or `wanderingLegends` with a `BLACK_THREAD` / grudge toward the player's sect!

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11. [BRAIN-CMB-10] Real-Time Tactical Combat Brain Contradictions (combat.js)

11.1 Exact Code Anchors

- js/combat.js:
  - fleeThreshold (lines 125–128) & flee roll (lines 374–383)
  - C.resolveDefensiveInterception bodyguard dash (lines 400–465)
  - C.resolveCognitiveArchetype (lines 567–584)
  - chooseSkill & C.slay Oath Violations (lines 630–710, 804–807)

11.2 Root-Cause Combat AI Bugs

Defect 10A — Non-Existent intellect.tactics Key + Fallback Misclassifies >60% of Monks, Tanks & Healers as 'CUNNING_ASSASSIN' (combat.js:574–584)

Inspect lines 574–584 of js/combat.js:

```javascript
const aggr = E(u, 'personality.aggression');
const inte = E(u, 'personality.integrity');
const tact = (u.h && u.h.intellect && u.h.intellect.tactics) || E(u,
'personality.rationality') || 50;
const path = u.cult && u.cult.path;
if (path === 'STRATEGY_GOVERNANCE' || path === 'ARRAY_TALISMAN' || tact >= 72)
return 'GRAND_TACTICIAN';
if (aggr >= 75 || path === 'DEMON') return (tact >= 58 ? 'CUNNING_ASSASSIN' :
'BLOODTHIRSTY_BERSERKER');
if (inte >= 68 || path === 'GOD' || path === 'XIAN') return 'HONORABLE_DUELLIST';
return 'CUNNING_ASSASSIN';
```

1. **u.h.intellect.tactics DOES NOT EXIST in human_core.js!** (intellect has planning and judgement, never tactics.)
2. Look at the final line: **return 'CUNNING_ASSASSIN'!**
   - Any BODY Tank, BUDDHA_DHARMA Monk, SAGE_SUPPORT Healer, ARENA_GLORY Champion, SHADOW_SAMURAI Bushido Duelist, CRAFT_CREATION Blacksmith, or BEAST_SOVEREIGN Tamer whose integrity is < 68, aggression < 75, and rationality < 72 (i.e., **the majority of normal Gaussian-generated NPCs!**) is assigned **'CUNNING_ASSASSIN'!**
   - In computeTargetPriorityScore (line 599), 'CUNNING_ASSASSIN' applies role === 'TANK' -> -25 and HEALER/CASTER -> +55, making

righteous Buddhist Monks and Heavy Body Tanks ignore frontline threats to dive backlines like assassins!

- **Fix**: Read `(E(u, 'intellect.planning') * 0.55 + E(u, 'intellect.judgement') * 0.45)` for tactical acumen; classify `BUDDHA_DHARMA`, `BODY`, `ARENA_GLORY`, `SHADOW_SAMURAI`, `ANCESTRAL_RITUAL` (with `inte >= 45`) as `'HONORABLE_DUELLIST'` (or `'GUARDIAN_VANGUARD'`); restrict `'CUNNING_ASSASSIN'` to stealth/assassin/poison paths (`NINPO_SHADOW`, `VENOM_GU`, `SOUL_PUPPETRY`, `VOID_HERETIC_PACT`, `DEMON`) or low-integrity (`< 42`) opportunists, and default neutral fighters to `'BALANCED_WARRIOR'`!

Defect 10B — Bitter Enemies Heroically Bodyblock Lethal Attacks for People They Hate (`combat.js:405–452`)

In `C.resolveDefensiveInterception`:

- When `d` is at `< 25% HP`, any nearby `TANK` on the same side (`sc = +1.8`) easily exceeds the `1.35` threshold **even if `SECT.rel(al, d.id).a === -95` (`bond === 'ศัตรู'`) or they share a `BLACK_THREAD` blood grudge**!
- Worse: line 449 immediately calls `bindThread(W, gUnit, d, 'GOLD_THREAD', ...)`—turning two bitter enemies into Golden-Thread life saviors because the Tank wasn't allowed to refuse sacrificing themselves for a hated rival!
- **Fix**: Subtract `Math.max(0, -(rel.a || 0)) / 25 + ((rel.v || 0) > 50 ? 1.2 : 0) + (hasBlackThread ? 3.0 : 0)` from `sc` (unless `E(al, 'personality.integrity') >= 82 && E(al, 'values.sectDuty') >= 80`, representing a rare paragon who protects even a hated sectmate out of pure duty!).

Defect 10C — Combat AI Forces `VOW_NO_KILL` Monks & `VOW_NO_RETREAT` Warriors to Break Their Own Vows (`combat.js:376, 646, 804`)

1. **`VOW_NO_KILL` / `VEGETARIAN_PACIFIST`**: In `C.applyDamage` (`line 365`), when attacker `a` holds an active `VOW_NO_KILL` / `VEGETARIAN_PACIFIST` oath (`a.daoOath?.upheld`) and is not berserk (`heartDemon < 70`), `a` should **pull the final blow against human foes** (`setHp(d, 1); d.yield = true;` -> subduing/capturing via `SECT.capturePrisoner`!) instead of slaying `d` and self-destructing their own Dao Oath in `C.slay`!
2. **`VOW_NO_RETREAT`**: In `fleeThreshold` (`line 126`), an NPC with active `VOW_NO_RETREAT` (or defending their spouse/child/apprentice on the same battlefield!) should have `fleeThreshold = 0` unless `fear > willpower + 25` (true mental collapse).

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12. [BRAIN-MKT-11] Hidden Dead-Ternary Typos & Missing Skill Keys in Evaluators (expedition.js, sect.js, bottleneck.js, external_market.js)

12.1 Smoking-Gun Dead Ternaries & Wrong Property Names Found in Live Code

File & Line	Exact Current Code	Bug Explanation	Exact Fix
js/expedition.js:44	e('social.diplomacy' in {} ? '' : 'personality.empathy')	'social.diplomacy' in {} is **always false**! diplomacy is in n.h.skills.diplomacy, so NPCs completely ignore their Diplomacy skill when volunteering for diplomatic missions!	`(E(n, 'personality.empathy') * 0.5 + (n.h.skills.diplomacy
js/expedition.js:217	HC.trainBody(v, 'vitality' in {} ? '' : 'endurance', 4)	'vitality' in {} is **always false**! Spirit Springs never train vitality as intended.	HC.trainBody(v, W.rng.chance(0.5) ? 'vitality' : 'endurance', 4)
js/expedition.js:239	E(a, 'social.teaching' in {} ? '' : 'interests.teaching')	'social.teaching' in {} is **always false**, ignoring social.teaching (which is a real HC attribute!) during talent scouting!	(E(a, 'social.teaching') * 0.6 + E(a, 'interests.teaching') * 0.4)
js/sect.js:76–78 (observableCapability)	sk.LIGHTFOOT, sk.HERB_LORE, sk.farming, and only 7 weapons	**LIGHTFOOT, HERB_LORE, and farming DO NOT EXIST in n.h.skills!** Actual keys are MOVEMENT	Replace dead keys with sk.MOVEMENT, sk.herbalism, and include all 16 weapon skills in martialSkill!

File & Line	Exact Current Code	Bug Explanation	Exact Fix
		and herbalism. Also omits **all 9 newer weapons** (ZITHER, BRUSH, WHIP, ROSARY, CLAW, NEEDLE, FAN, ASTROLABE, ABACUS)!	
js/sect.js:260 (SECT.relevance)	Math.max(sk.SWORD, sk.BLADE, sk.SPEAR, sk.FIST_PALM, sk.BOW, sk.STAFF, sk.FOCUS, sk.BODY_REFINING)	**Omits all 9 newer weapon skills** (ZITHER..ABACUS)! Masters of those 9 weapons get **martial = 0** when scored for martial, guard, vanguard, discipline, and warden Elder appointments!	Compute martial across all 16 weapon skills + BODY_REFINING + MOVEMENT!
js/external_market.js:465 (outsideTrade.can)	const v0 = vList[0], vStn = v0 ? v0.stones : 0;	Cache _otCache ignores vList[1]'s arrival/wallet and ignores changes in n.inv.mats / n.inv.pills for 30 ticks; memberQuote also cannot sell surplus n.inv.gear or buy CRYSTAL / SUB_MANUAL / RECIPE_SCROLL!	Hash all active traders' tradeCount + n._ver + bag counts, and allow personal gear/scroll/crystal trades!
js/bottleneck.js vs enterClosedDoorSeclusion	AI.ACTS.consolidate vs CULT.enterClosedDoorSeclusion	NPCs at major bottlenecks (stage >= 3, BN.decide === 'wait') never autonomously enter closedDoorSeclusion or invite their	Let cautious/patient Stage-3+ bottlenecked NPCs autonomously enter closedDoorSeclusion and pick their highest-trust

File & Line	Exact Current Code	Bug Explanation	Exact Fix
		`mentor`/`spouse`/`sworn friend` as `protectorId` unless the player toggles UI automation!	available bond (`rel.a >= 50` or `mentor`/`spouse`) as `protectorId`!

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13. Mandatory End-to-End Verification Suite Spec (For System #02 Part 3)

Add `test/v270_npc_brain_part3_executive_combat_lineage_suite.js` asserting:

1. **`test_autonomous_executive_enacts_charter_founds_peaks_and_convenes_council`**: Run a 120-day headless simulation with 0 player UI clicks under a high-commerce Sect Master and 25+ disciples; verify `SECT.masterDecision` / council autonomously updates Charter/Peaks/Outposts/Council agendas according to NPC personalities.
2. **`test_marriage_syncs_companion_and_mentorship_transmits_dying_qi`**: Trigger marriage via `AI.updateBond` (`spouse`) and verify `companionId` + `RED_THREAD` sync; assign a disciple via `SECT.takePersonalDisciple`, kill the master via `AI.kill`, and verify `transmitQi` fires for that disciple.
3. **`test_combat_archetypes_vows_and_enemy_bodyblock_refusal`**:
   - Verify a balanced `BUDDHA_DHARMA` monk resolves to `'HONORABLE_DUELLIST'` (not `'CUNNING_ASSASSIN'`).
   - Verify a `VOW_NO_KILL` monk subdues (`yield = true, hp = 1`) rather than slaying and breaking their oath.
   - Verify a Tank with `rel.a = -85` (`'ศัตรู'`) refuses `BODYGUARD_INTERCEPT` for their hated enemy.
4. **`test_sect_relevance_and_observable_capability_include_all_16_weapons_and_real_skill_keys`**: Give an NPC `CLAW = 90, MOVEMENT = 80, herbalism = 75`; verify `SECT.observableCapability` and `SECT.relevance('martial', npc)` score `> 60` instead of `0`.